

Particulate Oxidative Potential Associated with Fireworks Activity during Diwali at a site in the Indo-Gangetic Plain

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Research Article

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Abstract

The dithiothreitol (DTT) assay was used to investigate the oxidative potential (OP) of atmospheric fine particles ($PM_{2.5}$) to form reactive oxygen species (ROS) during Diwali festival firework activity at a site in the Indo-Gangetic Plain, India. During the study period, the 12-hour mean $PM_{2.5}$ was reported to be $263 \pm 151 \mu g m^{-3}$, which was 4.4 times higher than the NAAQS standards. Increased levels of gaseous pollutants (NO_x , SO_2 , CO and O_3), as well as metal concentrations such as Ba, Pb, Cu, Fe, Mg, K, Al, and Mn, were also observed during fireworks displays. The increased $PM_{2.5}$ - NO_x slope for fireworks, which included traffic emissions, revealed that fireworks played a significant role. The highest value of OP of PM was detected in the next morning of Diwali, with a volume-based DTT value of $1.4 nmol min^{-1} m^{-3}$ and a mass-based DTT value of $11.8 pmol min^{-1} \mu g^{-1}$, implying higher PM-related DTT activity due to fireworks. A positive association was found between redox-active metals like Cr, Cu, Ni, and V and DTT activity which could be due to the ability of these metals to catalyse ROS generation in ambient air. Ba and Se concentrations in the particulate matter, which are key components of firecrackers, were likewise highly linked to DTT activity. During the daytime, ozone levels were strongly correlated ($r^2 = 0.9$) with OPv activity; this may be due to photochemical processes which may lead to the formation of tropospheric O_3 . Elements C, O, Al, Si, S, K, and Ba were found to be the most prevalent in elemental mappings, which suggests that fireworks may have contributed to their abundance. A comparison of the daily OPv activity and hazard index (HI) suggests that the HI may be a poor metric for measuring the health impacts of PM exposure on human health.

Highlights

- This study focuses on the impact of oxidative properties of $PM_{2.5}$ during the Diwali period.
- DTT Assay was used to estimate the oxidative potential (OP) of $PM_{2.5}$.
- Transition metals in PM seem to have high oxidative activity linked with a wide range of health effects, including effects on the cardiovascular and respiratory systems.

1. Introduction

Fireworks are set off during festivals and celebrations namely Diwali in India (Prabhu et al. 2019; Singh and Srivastava 2020), Chinese Spring Festival in China (Pang et al. 2021), Independence Day in the USA (Jia et al. 2020), Lantern festival in Taiwan (Lin et al. 2014), Guy Fawkes in the UK (Godri et al. 2010) and New Year's celebrations in many nations (Retama et al. 2019; Rindelaub et al. 2021) which, causes short-term deterioration of air quality. Fireworks are composed of a variety of metal alloys, such as Sr, Ba, Na, Ti, Zr, Mg, Cu, and Al, as well as potentially harmful substances like KNO_3 , C, and S, to create the vibrant effects (Hickey et al. 2020; Rindelaub et al. 2021). Additionally, a few studies have also observed that the emission of organic pollutants occurs during fireworks (Jia et al. 2020). Short-term exposure to pollutants such trace gases, particulate matter (PM), and various metals like Al, Mn, Ni, Cr, Fe and Cd may result from the fireworks, posing substantial health concerns (Prabhu et al. 2019; Singh et al. 2019).

The impact of fireworks on poor air quality has been extensively documented in the literature (Prabhu et al. 2019; Singh et al. 2019; Hickey et al. 2020; Singh and Srivastava 2020). The effects of fireworks have been widely reported, including reduced visibility (Singh et al. 2019) and increased particulate load during the German New Year (Drewnick et al. 2006), Independence Day in the United States (Jia et al. 2020) and Spain (Moreno et al. 2007) and the Diwali celebration in India (Ambade 2018; Ganguly et al. 2019; Prabhu et al. 2019; Sahu et al. 2020; Singh and Srivastava

2020). Attri et al. (2001) observed a sudden increase in the concentration of O_3 in the next morning of fireworks from 9:00 p.m. to 2:00 a.m. in the air (Attri et al. 2001). Several studies in India have also found that fireworks may be a contributing factor in the deterioration of air quality during Diwali celebrations. In comparison to average winter days in December 1999, Ravindra et al. (2003) found that the concentration of SO_2 increased by about ten-fold during Diwali, while PM_{10} , TSP, and NO_2 levels increased by a factor of two. Kulshrestha, et al. (2004) observed that Sr, Al, K, and Ba concentrations increased by 15, 18, 25, and 1091 times, respectively, during Diwali compared to pre-Diwali. This study also found that Diwali firework displays are a massive source of air pollution, with exceptionally high amounts of metals in the ambient air (Kulshrestha et al. 2004). In comparison to a typical day, Barman et al. (2008) found that levels of SO_2 , NO_x , and PM_{10} spiked by a factor of 2, 2 and 3 times during the pre-Diwali period. Furthermore, except for Fe, trace elements such as Co, Cu, Ca, Cd, Cr, Mn, Pb, Ni, and Zn were greater in PM_{10} samples than on normal days and before Diwali (Barman et al. 2008). Sarkar et al. (2010) attributed 20 to 35% of total PM_{10} to firework emissions in Delhi, as well as a rise in Ca, Mn, Al, S, Na, Mg, Sr, K, Ba, and elemental carbon (EC) levels by a factor of 1.6, 2.7, 3.2, 4, 5, 5.8, 15, 18, 264, and 4.3 respectively, in comparison to normal values (Sarkar et al. 2010). A similar increase in PM_{10} levels was also observed in Jamshedpur (Ambade 2018). In comparison to pre-Diwali ($226 \mu g m^{-3}$), PM_{10} mass concentrations increased to $974 \mu g m^{-3}$ on Diwali and $787 \mu g m^{-3}$ on post-Diwali, and 55–73% of particles were recorded in the particle size less than $1 \mu m$ (PM_1) size range. Due to fireworks, levels of metals such as K, Al, S, Ba, and Cl increased dramatically during Diwali, accounting for 34% of the total metals examined, compared to pre-Diwali (1.0%) and post-Diwali (2.0%) (Singh and Srivastava 2020). However, in certain metropolitan regions, contamination remains in the ambient air for a short span, but fine and ultrafine particles stay for few days (Kulshrestha et al. 2004; Singh and Srivastava 2020). Asthmatic attacks, lung infection, eye hypersensitivity, cough, fever, and cardiovascular problems can all be triggered by an unanticipated increase in pollution levels during fireworks (Hirai et al. 2000). Few studies have addressed the potential negative health consequences of fireworks exposure in India, which found a 30 to 40% rise in incidences of respiratory illnesses, wheezing, and exacerbation of bronchial asthma and bronchitis in people of all ages (Liu et al. 2017). Not only at urban locations, but also at suburban and rural sites in China, fireworks have been shown to have antagonistic impacts with increase in $PM_{2.5}$ concentrations when compared to a regular day ((Ravindra et al. 2003; Zhang et al. 2017). Only a few studies have established the toxicity of fireworks PM on the respiratory system, despite the widespread usage of firecrackers to commemorate important occasions around the world. Recently, a striking relationship was discovered between particulate OP and trace metals associated with fireworks, implying that fireworks have a negative influence on human health. Following exposure to PM, reactive oxygen species are produced in numerous extracellular and cellular compartments, with evidence that distinct components (biological material, polycyclic aromatic hydrocarbons/ quinones, and trace metals) might cause oxidative stress via diverse mechanisms (Godri et al. 2010; Hickey et al. 2020). In a recent study, Hickey et al. assessed the production of reactive oxygen species (ROS) in bronchial epithelial airway (BEAS-2B) and human pulmonary microvascular endothelial (HPMEC-ST1.6R) cells exposed with size-fractionated fireworks particles. It was determined that various forms of fireworks were related with large increases in ROS in both cell types, and that particles produced by firecrackers can be harmful to mammalian cells and lungs (Hickey et al. 2020). To gain insight into the chemistry of particulate OP and trace metals associated with fireworks in the atmosphere, the DTT assay was used to study the short-term variation in oxidative properties of $PM_{2.5}$ during Diwali celebrations in the year 2019. Even though this celebration lasts only a few hours, it pollutes the environment and harms human health. This short (~ one week) measurement period provided us a peculiar opportunity to link the ambient $PM_{2.5}$ to its OP, due to firework activity.

2. Materials And Methods

2.1. Site description, Chemical Analysis

PM_{2.5} samples were collected at a traffic and residential site (Bhagwan Talkies Crossing, adjacent to Agra-Delhi Highway NH-19) in Agra, India (Fig. 1). Agra (27° 10' 47" N, 78° 01' 02" E; 169 MSL) situated in Uttar Pradesh lies in the Indo-Gangetic Plain. It has an area of 188 km² and is one of the most populated cities in Uttar Pradesh with a population of nearly 2.0 million. Four major national highways (NHs) and two expressways (EWs) pass through the city namely Agra-Jaipur Highway NH-21 Agra-Gwalior Highway NH-44, Agra-Aligarh Highway NH-509, Agra-Delhi Highway NH-19, Yamuna expressway (Delhi-Agra) and Agra-Lucknow expressway. Among all, NH-19 is one of the busiest highways, which connects Delhi to Kolkata via Agra. The sampling site, Bhagwan Talkies Crossing is one of the busiest crossings on NH-19 with residential colonies lying sideways and high level of vehicular pollution, is caused by the highway and localized traffic congestions (RITES 2017).

PM_{2.5} samples were collected on 47-mm quartz fibre filters (Pallflex) using a fine particulate sampler (Envirotech, APM 550), operated at a flow rate of 16.6 L min⁻¹. For the duration of the study, hourly concentrations of trace gases such as CO, NO₂, SO₂, O₃ and PM_{2.5}, as well as the meteorological parameters; temperature (T), relative humidity (RH), wind speed (WS), rainfall (RF), solar radiation (SR), and wind direction (WD) were obtained from the Central Pollution Control Board's online portal (from 24-10-2019 to 31-10-2019; Diwali period). The sampling period from October 24 to October 26 was referred to as pre-Diwali, October 27 to October 28, as Diwali, and October 29 to October 31, as post-Diwali. The sampling was done twice a day, from 8:00 a.m. to 7:55 p.m. and from 8:00 p.m. to 7:55 a.m., the next morning. Before sampling, the filter paper was cleaned by heating it at 800°C for 7 hours to eliminate volatile contaminants, then conditioned for 48 hours in a desiccator at 25°C and 40% relative humidity, and weighed using a microbalance (Mettler Toledo, ME 204). To prevent contamination and deterioration of the sample, the loaded filter papers were removed using forceps and placed in the cassette, which was then stored in low-density polyethylene (LDPE) zip-lock bag at -20°C until analysis. On Diwali days, air quality was assessed and compared to typical days (pre-Diwali and post-Diwali days).

One-fourth of the exposed filter was digested in 10 ml nitric acid (69%, Merck) on a hot plate for 2 h at 90°C for metal analysis. The digested solution was filtered using a microporous membrane filter (Sartorius 393) after cooling, and the final volume was made up to 40 ml in a PTFE bottle with de-ionized water (resistivity = 18.2 M/cm). Then, using an inductively coupled plasma-optical emission spectrometer, these acidic aqueous samples were examined for sixteen metals (Ba, Ca, Cr, Na, Ni, Cu, V, Pb, Fe, Mg, Zn, Cd, K, Al, Se, Mn) (ICP-OES, Agilent 5110, USA). Table S1 lists the ICP-OES operating parameters. The equipment was calibrated with the use of a multi-element standard (Agilent, USA). For quantification, a seven-point standard curve was produced with concentrations ranging from 50 to 25000 g/L (50 g/L, 100 g/L, 500 g/L, 1000 g/L, 5000 g/L, 10000 g/L, and 25000 g/L). All of the standard curves for specific metals had linear regression coefficients better than 0.95 ($r > 0.95$). As a blank and for dilution of standards, 5% HNO₃ (v/v) was used. All samples were analysed in triplicate, with the concentration estimated as the average of the triplicates. Field blank filters were also treated and analysed by the same procedure. The limit of detection (LOD) was determined by taking three times the standard deviation of blank concentration. LODs of metals were listed in Table S2. Then, the metal concentrations in the field blank were deduced from the concentration of the samples for all the analyses.

2.2. Measurement of Oxidative potential (OP) – The Dithiothreitol (DTT) Assay

The DTT assay was used to determine the oxidative potential of water-soluble PM_{2.5}. One-fourth of loaded quartz filters were extracted in de-ionized water by ultrasonication (model: USB 3.5 L H DTC, PCi Analytics) for 45 minutes (thrice, 15 minute each). Subsequently, insoluble suspended particles were removed from the extract using a microporous membrane filter (Sartorius 393), and the filtered extracts were incubated in an amber-coloured glass vial at 37 °C with 100 µM DTT (≥ 98%, SRL Chem) prepared in 0.1 M phosphate buffer solution (Thermo Fisher, pH 7.4). These vials were continuously shaken in an incubator shaker (Waiometra, India). A known aliquot of the reaction mixture was removed from the vial at various time intervals (0, 10, 20, and 30 minutes) and quenched with 10% (w/v) trichloroacetic acid (TCA, Thermo Fisher), 0.4M Tris-HCl (LOBA chemie, 20mM EDTA, pH 8.9), and 10 mM 5, 5'-dithiobis-2-nitrobenzoic acid (DTNB, SRL Chem). DTNB is transformed to 2-nitro-5-thiobenzoic acid (TNB), a yellow-coloured compound that is stable for two hours at room temperature (Charrier and Anastasio 2012). The concentration of TNB is proportional to the concentration of unreacted DTT. Subsequently, it was measured by UV-VIS spectrophotometer (UV-1800, Shimadzu) at 412 nm. The OP response was evaluated as the depletion rate of unreacted DTT, according to the experimental procedure described in the literature (Charrier and Anastasio 2012; Patel and Rastogi 2018a). The depletion rates were calculated using seven experimental points and linear fitting of the unreacted DTT concentration as a function of time (at 0, 10, 20, 30, 40, 50 and 60 min). Several blanks (n = 10) were also analysed in parallel with the samples, and the blank value was subtracted from the DTT activity measured. To ensure data quality, all samples and blanks were run in duplicate, and samples were not accepted unless the relative standard deviation (RSD) values of duplicates were within 5%. In addition, four out of ten samples were repeated on various days; these measurements were consistent within 5–10%. The method was validated by measuring the DTT consumption rate for a certain standard and comparing it with the documented value (Charrier and Anastasio, 2012). The consumption rate of DTT (100 M starting concentration) was tested in the presence of 0.05 M 9, 10-Phenanthrenequinone (PQN) and found to be $0.68 \pm 0.16 \text{ M DTT min}^{-1}$. For the same amount of PQN, the measured value is identical to that reported by Charrier and Anastasio (2012).

The OP was reported in terms of DTT consumption rate per cubic meter of total ambient air volume (OP_v) as Eq. 1 and DTT consumption rate per microgram of total PM mass (OP_m) collected on the sampled filter as Eq. 2 (Fang et al. 2017).

$$OP_v \left(nmolmin^{-1}m^{-3} \right) = \frac{r_s \left(nmolmin^{-1} \right) - r_b \left(nmolmin^{-1} \right)}{V_t \left(m^3 \right) \times A \times V}$$

1

$$OP_m \left(pmolmin^{-1}\mu g^{-1} \right) = \frac{r_s \left(pmolmin^{-1} \right) - r_b \left(pmolmin^{-1} \right)}{M_t \left(\mu g \right) \times A \times V}$$

2

where r_s and r_b are the DTT consumption rates of sample and blank, respectively. V_t and M_t are the total volume and the total particle mass of sampled air (subtracting the volume and mass of the blank filter respectively), respectively. A is the ratio of the filter paper covered with sample (A_h) and total filter paper (A_t) respectively. V is the ratio of the sample volume participating in reaction (V_s) and extraction volume (V_e) respectively. In general, the OP_v response is an extensive parameter depending on the PM_{2.5} mass concentration, while the intrinsic OP_m represents the contributions of specific sources to the overall PM oxidative potential (Romano et al. 2020).

2.3. SEM/EDS Analysis

The morphology, elemental composition, and optical properties of the collected particles were analysed using Field Emission Scanning Electron Microscope (FESEM, JSM-7610F Plus, JEOL) equipped with Energy Dispersive Spectrometer (EDS, Oxford Instruments) through AZTEC software. To capture an overall perspective of the PM_{2.5} during the Diwali period, the loaded filters were mapped, and further individual analyses were performed on specific particles of interest. The sampled filter papers were randomly cut into the size of about 1 mm² and mounted on electron microprobe stubs with the help of carbon-tape. In order to achieve maximum conductivity, the mounted filters were sputter coated with platinum in an auto-fine platinum coater (JEC-3000 FC, JEOL) and then placed in the SEM chamber for analysis. The working conditions were set at 5 to 15 kV accelerating voltage with a magnification power of 10000X.

2.4. Data Quality Assurance and Quality Control (QA/QC)

For quality assurance (QA) and quality control (QC), the field blank, reagent blank and certified reference materials (CRM) were analysed with other samples for each type of chemical analysis. During the study period, one field blank filter extract along with de-ionized water blank was used as the negative controls for PM_{2.5} samples (n = 10) analysed. PQN known to be sensitive for the DTT assay, was used as the positive control and was analysed with PM_{2.5} samples to ensure the stability of the procedure. The recovery efficiencies of sixteen metals ranged between 65–106%. The concentration of the field blank was less than 10% of that of the samples on average throughout all sample analyses. The extraction procedure's repeatability and reliability were demonstrated by the recovery rates. Glassware were thoroughly cleaned in acid and dried in the oven to avoid contamination.

2.5. Statistical analysis

The data was analysed using descriptive statistics, and the mean and standard deviation were calculated. The monotonic associations between the OP and each component identified in PM samples and trace gases were assessed using Pearson's correlation coefficients, which were examined using SPSS ver. 16 software (IBM Corp., USA). A significant level was defined as a p value less than 0.05.

3. Results And Discussion

3.1. Diurnal variations of trace gases

The diurnal variations of trace gases measured at the present site are shown in Fig. 2. The 12-h mean concentrations of NO_x, SO₂, CO and O₃ were 70.3 ± 26.2 µg m⁻³ (range : 40.8 to 116.3 µg m⁻³), 15.8 ± 2.2 µg m⁻³ (range : 11.6 to 20.0 µg m⁻³), 1.4 ± 0.4 mg m⁻³ (range : 0.7 to 2.1 mg m⁻³) and 26.3 ± 13.0 µg m⁻³ (range : 12.6 to 58.5 µg m⁻³), respectively during study period which did not exceed the levels that are recommended by NAAQS (NAAQS 2009), indicating no threat to human health (Table 1). Day-time (8:00–19:55 hours) and night-time (20:00–7:55 hours in the next morning) mean values of NO_x, SO₂, CO and O₃ during pre-Diwali, Diwali and post-Diwali are listed in Table 2. For NO_x, a bimodal distribution with an early morning peak at 2:00–3:00 hours and a night peak at 18:00–24:00 hours was observed, which may be due to the combination of boundary layer processes and anthropogenic emissions including firework chemistry. However, low values (~ 30 µg m⁻³) were observed during the afternoon hours (Fig. 2 (a)). The maximum volume of precursors (i.e., NO_x, CO, VOCs) is necessary to sustain high ozone levels in the day-time, which could be the probable reason for the decrease in NO_x levels during day-time. During the day, NO_x also play a vital role in the oxidizing capacity by partitioning, formation and loss of free radicals (OH[•], HO₂[•] and RO₂[•]) (Gaur et al. 2014). The results also revealed that SO₂ levels peaked at 2:00 hours in the study

period, which were lower than the NAAQ safety standards ($80 \mu\text{g m}^{-3}$ for a 24-hour average), owing to traffic-related emissions from commercial vehicles ($26.1 \mu\text{g m}^{-3}$ a day before Diwali, $24.8 \mu\text{g m}^{-3}$ on Diwali day, and $27.7 \mu\text{g m}^{-3}$ a day after Diwali) (Yerramsetti et al. 2013). Furthermore, it began to increase between 11:00 and 18:00 hours, indicating possible traffic emissions, as Bhagwan Talkies Crossing is a traffic-dominated site with local industrial operations such as ferrous and non-ferrous metal casting, tanneries, and so on (Fig. 2 (b)) (Sah et al. 2019). However, the evening peak continued till 22:00 hours on Diwali day. An early morning peak at 2:00–3:00 hours during all three periods was observed for NO_x and SO_2 , which is due to the exhaust emitted from the heavy-duty diesel vehicles (HDVs) namely truck multi axle (3 and above) and freight trucks in late night which pass from the NH-19 (adjacent to the sampling site) (RITES 2017). Concentrations of CO were also found to be lower (day: 0.8 ± 0.2 and night: $1.3 \pm 0.3 \text{ mg m}^{-3}$) during the pre-Diwali (day: 1.3 ± 0.1 and night: $1.7 \pm 0.2 \text{ mg m}^{-3}$) and post-Diwali (day: 1.6 ± 0.4 and night: $1.8 \pm 0.2 \text{ mg m}^{-3}$) periods (Table 2). When compared to pre-Diwali, CO levels increased by 63% during the day and 31% at night on Diwali day, and similar increases have been documented in previous studies (Gaur et al. 2014; Parkhi et al. 2016). Due to heavy traffic, vehicular emissions are always consistent throughout the day, which can modulate CO levels by the boundary layer evolution. During the night, however, the height of the boundary layer decreases, resulting in CO confinement at the surface (Gaur et al. 2014), causing high CO concentration in the late evening (17:00 to 22:00 hours) (Fig. 2(c)). On Diwali, the concentration of O_3 was $49.9 \pm 12.2 \mu\text{g m}^{-3}$, followed by post-Diwali ($35.9 \pm 4.6 \mu\text{g m}^{-3}$) and pre-Diwali ($28.6 \pm 4.6 \mu\text{g m}^{-3}$) in day-time (Table 2), which is thought to be caused by photo-dissociation of molecular O_2 in the presence of ultraviolet light. A rise in ozone concentration was observed between 10:00 to 18:00 hours whereas the minimum appeared at night. Surface level O_3 is not a direct product of fireworks, it increases due to a rise in the level of its precursors gases, volatile organic compounds (VOCs) and carbon monoxide (CO), which is catalysed by the $\text{OH}^\bullet$ and regulated by NO_x , which were emitted from firework activities that took place in the previous night. The decline in night-time O_3 concentrations could be due to the titration of O_3 by surface emissions of NO and ground-level destruction of O_3 in the shallow boundary layer. It was also noted that the broad peak of O_3 concentration was observed during post-Diwali in the day-time (Fig. 2 (d)), which may be attributed to the presence of chemical species associated with fireworks and is responsible for the formation of tropospheric O_3 during day-time (Attri et al. 2001; Venkatachari et al. 2007).

Table 1
Comparison of Ambient pollutant concentrations of normal, pre- Diwali and post-Diwali days to Diwali day with similar firework studies

Event	City, Country	Study period	Ambient pollutant concentration ($\mu\text{g m}^{-3}$)						DTT Activity
			PM _{2.5}	PM ₁₀	O ₃	CO	NO _x	SO ₂	OPv (nmol min ⁻¹ m ⁻³)/ OPm (pmol min ⁻¹ μg^{-1})
^a Guy Fawkes Night (Godri et al. 2010)	Marylebone, UK	2007	44.3 $\pm$ 17.6	-	-	-	83.3 $\pm$ 3.5	23.6 $\pm$ 4.0	6.5 $\pm$ 1.5 (OPm)
^b Chinese Spring Festival (Huang et al. 2012)	Shanghai, China	2009	775	> 900	-	-	286	438	-
^c Commonwealth Games (Beig et al. 2013)	New Delhi, India	2010	> 2000	> 2500	-	-	-	-	-
^d Diwali (Saha et al. 2014)	Kolkata, India	2013	-	875	-	-	190 (NO ₂)	125.21	-
^e Guy Fawkes Night (Singh et al. 2015)	Nottingham, UK	2000–2012	31.2–38.7	37.4	-	-	-	-	-
^f Diwali (Pervez et al. 2016)	Bhilai, India	2012	1501.2	-	-	-	-	-	-
^g Diwali (Parkhi et al. 2016)	New Delhi and NCR, India	2010 and 2011	1620 in 2010, 390 in 2011	2070 in 2010, 600 in 2011	> 50 ppbV	5–10 ppmV	-	-	-
^h Chinese Spring Festival (Zhang et al., 2017)	Beijing, China	2015	248.9	-	-	-	-	57.8	-
ⁱ Diwali (Ambade 2018)	Jamshedpur, India	2014	-	500.5	-	-	73.4	8.6	-
^j Diwali (Prabhu et al. 2019)	Dehradun, India	2017 and 2018	79.92–324.97	57.35–256.12	-	-	-	-	-

Event	City, Country	Study period	Ambient pollutant concentration ($\mu\text{g m}^{-3}$)						DTT Activity
			PM _{2.5}	PM ₁₀	O ₃	CO	NO _x	SO ₂	OPv (nmol min ⁻¹ m ⁻³)/ OPm (pmol min ⁻¹ μg^{-1})
^k Guy Fawkes Night (Rindelaub et al. 2021)	Auckland, New Zealand	2019	-	35.8	-	-	-	-	-
^l Diwali (Singh et al. 2020)	Delhi, India	2018	-	974.55	-	-	-	-	-
^m Diwali (Present study)	Agra, India	2019	263 $\pm$ 151	-	26.3 $\pm$ 13.0	1.4 $\pm$ 0.4	70.3 $\pm$ 26.2	15.8 $\pm$ 2.2	0.35 $\pm$ 0.45 (OPv) and 6.1 $\pm$ 4.2 (OPm)

Table 2

Concentrations of particulate matter and trace gases was taken by CPCB portal on pre-Diwali, Diwali and post-Diwali day

Pollutants	Concentration					
	pre-Diwali		Diwali		post-Diwali	
	Day	Night	Day	Night	Day	Night
PM _{2.5} (μgm^{-3})	52.7 $\pm$ 17.1	65.6 $\pm$ 9.2	63.0 $\pm$ 9.6	116.3 $\pm$ 23.6	97.6 $\pm$ 35.0	97.6 $\pm$ 15.9
NO _x (μgm^{-3})	41.6 $\pm$ 2.4	78.9 $\pm$ 13.4	51.1 $\pm$ 5.8	108.1 $\pm$ 11.6	50.9 $\pm$ 6.2	97.6 $\pm$ 6.0
SO ₂ (μgm^{-3})	14.1 $\pm$ 3.3	16.9 $\pm$ 3.1	14.7 $\pm$ 0.4	15.9 $\pm$ 1.7	16.2 $\pm$ 1.7	16.6 $\pm$ 1.4
CO (mgm ⁻³)	0.8 $\pm$ 0.2	1.3 $\pm$ 0.3	1.3 $\pm$ 0.1	1.7 $\pm$ 0.2	1.6 $\pm$ 0.4	1.8 $\pm$ 0.2
Ozone (μgm^{-3})	28.6 $\pm$ 4.6	16.9 $\pm$ 4.3	49.9 $\pm$ 12.2	15.4 $\pm$ 1.7	35.9 $\pm$ 4.6	15.1 $\pm$ 0.2

3.2. Characteristics of PM_{2.5}

During the study period, the 12-hour mean PM_{2.5} was found to be 263 $\pm$ 151 $\mu\text{g m}^{-3}$ (Table 1). Table S3 shows PM_{2.5} concentration trends and meteorological conditions during the study period, both during the day and at night. The 12-hour mean PM_{2.5} (263 $\mu\text{g m}^{-3}$) was nearly 4.4 times higher than India's National Ambient Air Quality Standards (60 $\mu\text{g m}^{-3}$) (NAAQS 2009). A day before Diwali (26/10/2019), the mass concentration of PM_{2.5} was 108 and 153 $\mu\text{g m}^{-3}$ in the daytime and night-time, respectively, however on Diwali day (27/10/2019), the mass concentration of

PM_{2.5} was 131 and 412 $\mu\text{g m}^{-3}$ in the daytime and night-time, respectively. On the contrary, a day after Diwali (28/10/2019), particle concentrations peaked at 444 and 508 $\mu\text{g m}^{-3}$ during the day and night, respectively (Fig. 3). The increase could be due to the fireworks and other anthropogenic activities. Table 1 shows a comparison of pollutant concentrations during fireworks at different places across the globe. During the Diwali season, the PM_{2.5} mass concentration (263 $\mu\text{g m}^{-3}$) at this site was equivalent to that recorded during the Chinese spring festival in Beijing, China (249 $\mu\text{g m}^{-3}$, Zhang et al., 2017). PM_{2.5} mass concentrations at the present site were higher than those reported at other sites around the world, including Marylebone ($44 \pm 18 \mu\text{g m}^{-3}$, Godri et al., 2010) and Nottingham ($\mu\text{g m}^{-3}$, Singh et al., 2015) in the United Kingdom, and Shanghai, China (775 $\mu\text{g m}^{-3}$, Huang et al., 2012) and Bhilai (1501.2 $\mu\text{g m}^{-3}$, Pervez et al., 2016). During Diwali, similar increases in PM_{2.5} levels were seen in Delhi (Parkhi et al. 2016) and Dehradun (Prabhu et al. 2019). The PM_{2.5} mass concentration was also found to be higher at night than during the day, owing to the increased firework activity at night (Fig. 3). Aside from the use of firecrackers, meteorological conditions may have a role in the delayed dispersion of pollutants (Ganguly et al. 2019). Low relative humidity (42%) was observed during the day, which increased to 63% at night, while wind speeds ranged from 0.9 to 1.2 m/s during the day and 0.4 to 0.8 m/s at night, resulting in a decrease in boundary layer height, making the atmosphere less conducive to pollutant dispersion and dilution during the night. During the period, the prevailing wind direction was West–North West (W– NW) (Table S3).

Hourly concentrations of PM_{2.5} were also retrieved from CPCB (<https://app.cpcbcr.com/ccr/#/caaqm-dashboard-all/caaqm-landing>) to study the diurnal profile of PM_{2.5}. This is shown in Fig. 2 (e). The fireworks activity increased the concentrations of PM_{2.5} by 1.7- and 2.3-times during Diwali and after Diwali nights with respect to the night before Diwali, respectively, considering these from the sunset (19:55 hours) of the previous day until sunrise in the next morning (8:00 hours). Two peaks were observed with the particles concentration near or above 100 $\mu\text{g m}^{-3}$, as shown in Fig. 2 (e). Morning PM_{2.5} levels at 7:00–10:00 hours, is related to the secondary production of organic and inorganic particles 3–4 h after sunrise. The night levels occurs during the firework's pinnacle (Retama et al. 2019). On Diwali day (292 $\mu\text{g m}^{-3}$ at 24:00 hours) and the day after Diwali (240.5 $\mu\text{g m}^{-3}$ at 23:00 hours), the 24-hour standard of 60 $\mu\text{g m}^{-3}$ was significantly exceeded, whereas the day before Diwali (101 $\mu\text{g m}^{-3}$ at 24:00 h) the concentration was significantly less. These hourly peaks, as well as the ensuing 12-hour mean concentrations, were lower than those seen in China and India during huge fireworks displays (Huang et al. 2012; Kong et al. 2015; Parkhi et al. 2016; Pervez et al. 2016).

Increased emissions of metal oxides, metal salts, and other inorganic species may result from fireworks action late at night. Among these species, Ba has the highest contribution as a tracer of fireworks (Lin 2016), which could be due to the widespread use of various Ba salts in firecrackers, such as barium sulphate (BaSO₄), barium oxalate (BaC₂O₄), barium carbonate (BaCO₃), barium nitrate (Ba(NO₃)₂), and barium chlorate (Ba(ClO₃)₂), to impart green colour to the flashes (Kulshrestha et al. 2004). Furthermore, salts of Al (white), Na (yellow), Cu (blue), and Mg (bright white) help to create a variety of colours in firecrackers (Moreno et al. 2007). Fe, on the other hand, is essential for the production of sparks, while Ca improves the intensity of colour produced by other elements. Similarly, few other metals are utilised to create colour and sparkle in firecrackers.

Figure 4a illustrates the variation of trace metals such as Ba, Cd, Cr, Cu, Mn, Ni, Pb, Se, V, and Zn, whereas, the variation of crustal metals such as Al, Ca, Fe, K, Mg, and Na are represented in Fig. 4b. Among all the sixteen metals, Fe, K, Mg and Na were identified as the major metals with concentrations above 50 ng m^{-3} . Al, Ba, Ca, Cu and Zn varied from 1 to 50 ng m^{-3} while Cd, Cr, Mn, Ni, Pb, Se and V were mostly between 0.1 and 1 ng m^{-3} . V had the

lowest concentration ($< 0.1 \text{ ng m}^{-3}$). The concentration of metals was found in the order of $\text{Na} > \text{Fe} > \text{Mg} > \text{K} > \text{Ca} > \text{Zn} > \text{Ba} > \text{Al} > \text{Pb} > \text{Cu} > \text{Cr} > \text{Ni} > \text{Se} > \text{Mn} > \text{Cd} > \text{V}$. Ba, Ca, Cr, Ni, Se, Cu, V, Mg, Zn, and Cd were found to be 5.0, 2.7, 2.4, 2.0, 2.0, 2.0, 1.8, 1.6, 1.5, and 1.4 times higher, whereas Mn and K were moderately higher than the levels observed during pre-Diwali. The presence of various salts of colour-producing metals may thus account for the high concentration of these metals on Diwali day (Hickey et al. 2020; Rindelaub et al. 2021). It was also discovered that metal levels were low throughout the pre-Diwali and post-Diwali periods, implying that their retention duration was brief and that other local sources played a minor role (Kulshrestha et al. 2004). The tracers of fireworks (Al, Ba, Mg, Cu and K) were not significantly enriched as compared to their enrichment reported in earlier studies (Tanda et al. 2019). In India, in the year 2017, the Supreme Court banned the use of antimony, lithium, arsenic, lead and mercury in firecrackers and in the next year, the use of Ba was also banned in firecrackers and green crackers which contain lower amounts salts of Al, Ba, KNO_3 , and C were promoted to ensure release of minimum amounts of hazardous material into the atmosphere (Ramesh et al. 2021). Previous investigations in China (Kong et al. 2015), Taiwan (Tsai et al. 2012), Italy (Vecchi et al. 2008), Spain (Moreno et al. 2007), and India (Kulshrestha et al. 2004; Barman et al. 2008; Perrino et al. 2011) revealed similar levels of Al, Ba, Mg, and other metals.

A regression analysis between NO_x and $\text{PM}_{2.5}$ mass concentration was undertaken to provide a more qualitative assessment of the impact of fireworks in fine particles in the atmosphere. The data was divided into two period. The first reflected traffic and fireworks emissions (T + F) period, whereas the second represented traffic emissions (T) period alone (including vehicular tailpipe emissions, road dust re-suspension, and tyre and brake pad wear). T period was calculated using two weeks without firework activity: one week before Diwali and one week after Diwali (October 17–23, 2019 and November 1–7, 2019). T + F period, were assessed throughout Diwali week (October 24–31, 2019). Figure 5 shows the $\text{PM}_{2.5}$ - NO_x association for T period (slope = 0.1; y-intercept = $56.3 \text{ } \mu\text{g m}^{-3}$; $r^2 = 0.003$; $p > 0.05$; $N = 337$) and T + F period (slope = 0.7; y-intercept = $28.1 \text{ } \mu\text{g m}^{-3}$; $r^2 = 0.113$; $p > 0.05$; $N = 193$). The higher $\text{PM}_{2.5}$ - NO_x slope for T + F period, was noticed, which can be attributed to fireworks emitting more pollutants than biomass burning or other combustion sources (Godri et al. 2010). The meteorological parameters, namely SR, RH, AT, and WD remained similar during the two periods. The wind speed was 0.8 m/sec during T + F period and increased to 1.3 m/sec in T period, which would have promoted dispersal of pollutants in the T period.

Since, PM concentrations are closely related to wind speed and direction, a backward air mass trajectory study can also aid in pinpointing the source and pathway of PM. The 72-hour backward air mass trajectories in Agra were simulated using the National Oceanic and Atmospheric Administration (NOAA) Hybrid Single-Particle Lagrangian Integrated Trajectory (HYSPLIT) model (Stein et al. 2015) at 12:00 h (UTC) throughout the Diwali period and are shown in Figure S1. The trajectories were short which showed the firework effect is localized within a limited geographic scale during Diwali days from 26/10/2019 to 28/10/2019, suggesting potential need for local monitoring and control programs.

3.3. Oxidative potential (OP) of $\text{PM}_{2.5}$

The oxidative potential during the Diwali period was measured as OPv and OPm of $\text{PM}_{2.5}$ (Fig. 6). During the study period, 12-h mean volume-based OP values were found to be $0.35 \pm 0.45 \text{ nmol min}^{-1} \text{ m}^{-3}$. During Diwali period, the mean OPv value was highest ($0.77 \pm 0.44 \text{ nmol min}^{-1} \text{ m}^{-3}$) followed by pre-Diwali ($0.08 \pm 0.06 \text{ nmol min}^{-1} \text{ m}^{-3}$) and post-Diwali ($0.02 \pm 0.01 \text{ nmol min}^{-1} \text{ m}^{-3}$) period. The OPv measured on the night of Diwali, showed an elevated value ($0.85 \text{ nmol min}^{-1} \text{ m}^{-3}$), which was approximately 2.4 times of the average OPv obtained during the entire sampling period. This could be due to the overall increase in the ambient $\text{PM}_{2.5}$ concentration emitted from firecrackers. The mean OPv levels found at this location were compared to those found at other sites throughout the

world and in India. OPv obtained over the entire sampling period was $0.35 \pm 0.45 \text{ nmol min}^{-1} \text{ m}^{-3}$ (Table 1), which is in the typical range of the OP levels that have been observed over the southern United States; $0.1\text{--}1.5 \text{ nmol min}^{-1} \text{ m}^{-3}$ (Verma et al. 2014), Atlanta; $0.2\text{--}0.4 \text{ nmol min}^{-1} \text{ m}^{-3}$ (Gao et al. 2017), Lecce; $0.40 \pm 0.26 \text{ nmol min}^{-1} \text{ m}^{-3}$ (Chirizzi et al. 2017) and Xi'an; $0.19\text{--}1.10 \text{ nmol min}^{-1} \text{ m}^{-3}$ (Wang et al. 2020). OPv levels at the present site were lower than those measured at other sites around the world, including Indo-Gangetic Plain; $3.8 \pm 1.4 \text{ nmol min}^{-1} \text{ m}^{-3}$ (Patel and Rastogi 2018b), Beijing; $12.26 \pm 6.82 \text{ nmol min}^{-1} \text{ m}^{-3}$ (Yu et al. 2019) and Wuhan; $1.8\text{--}8.2 \text{ nmol min}^{-1} \text{ m}^{-3}$ (Liu et al. 2020), but higher than those measured at the urban site, USA; $0.04\text{--}0.18 \text{ nmol min}^{-1} \text{ m}^{-3}$ in previous studies (Xiong et al. 2017; Yu et al. 2018). Similarly, for the entire sampling period, 12-h mean OPm values were found to be $6.1 \pm 4.2 \text{ pmol min}^{-1} \mu\text{g}^{-1}$ (Table 1). The present site has a higher OPm value ($6.1 \pm 4.2 \text{ pmol min}^{-1} \mu\text{g}^{-1}$) than other urban sites, such as the southern United States; $0.005\text{--}0.1 \text{ nmol min}^{-1} \mu\text{g}^{-1}$ (Verma et al. 2014), Beijing; $0.13 \pm 0.1 \text{ nmol min}^{-1} \mu\text{g}^{-1}$ (Yu et al. 2019), and is comparable to levels observed during the Guy Fawkes Nights on fireworks event at Marylebone ($6.5 \pm 1.5 \text{ pmol min}^{-1} \mu\text{g}^{-1}$). OPm values at this site were lower than the values observed at Wuhan; $18.2\text{--}52 \text{ nmol min}^{-1} \mu\text{g}^{-1}$ on normal days (Liu et al. 2020).

Both during the pre-Diwali and post-Diwali periods, the OPv values were found to be higher at night than during the day, indicating the presence of numerous DTT-active components. A similar pattern was also seen in the OPm values. On the day following Diwali, on the other hand, OPv ($1.37 \text{ nmol min}^{-1} \text{ m}^{-3}$) and OPm values ($11.77 \text{ pmol min}^{-1} \mu\text{g}^{-1}$) were found to be higher in the daytime as opposed to the night-time (OPv value: $0.48 \text{ nmol min}^{-1} \text{ m}^{-3}$ and OPm value: $9.75 \text{ pmol min}^{-1} \mu\text{g}^{-1}$). The probable reason behind this could be the high concentrations of secondary pollutants (when the primary VOCs are oxidized by ozone, UV or reactive radicals) (Stevanovic et al. 2019) as well as good association with PM constituents (metals, organic compounds etc.) which were retained in the atmosphere (Molina et al. 2020).

To elucidate the relationship between OP and individual trace gases (NO_2 , SO_2 , CO , O_3), a linear regression analysis was performed. Table 3 shows the correlation coefficient (r) between OPv ($\text{nmol min}^{-1} \text{ m}^{-3}$) and the trace gases (NO_2 , SO_2 , CO , O_3) ($\mu\text{g m}^{-3}$). Strong correlations ($r^2 = 0.87$) were observed with ozone during day-time, which functions both as an atmospheric oxidant and as a precursor to other important oxidants such as hydroxyl and nitrate radicals formed during photochemical processes and/or ozonolysis reactions (Brown et al. 2020). Another important factor could be the presence of chemical species linked to fireworks that are responsible for tropospheric O_3 production (Attri et al. 2001; Xu et al. 2018). However, there was a weak correlation at night, most likely due to the lack of photochemically produced O_3 , implying that the intensity of the photochemical reaction was a significant factor influencing the generation of particulate ROS during the day (Hung and Wang 2001; Venkatachari et al. 2007). During the night, NO_2 had a good correlation with OPv ($r^2 = 0.42$), which could be related to the existence of nitrate radical, which is formed during the night by the reaction of NO_2 with ozone (Kameda et al. 2016). It's worth noting that a strong association between CO and OPv ($r^2 = 0.47$) was also detected during the day. These findings could indicate that high-level organics in both the gas and particle phases of vehicular emissions are responsible for the generation of PM-bound ROS, as organics are oxidised via hydroxyl radical-initiated processes (Brown et al. 2020).

Table 3
Correlation coefficients (r) of OPv with trace gases during Diwali period

Trace Gases	OPv	
	r (Day-time)	r (Night-time)
NO ₂	0.28	0.65
SO ₂	0.32	0.32
CO	0.68	0.23
O ₃	0.92	0.49

Additionally, Pearson's correlation coefficients were also calculated to find associations between OP and specific components in PM_{2.5} that contributed to DTT. The correlation matrix of the sixteen metals and OP is presented in **Table S4**. Strong correlations ($r > 0.70$, $p < 0.01$) were observed between OPv and Ba, Cd, Cr, Cu, Ni, Se and V, which are tracers of several anthropogenic activities. Metals like Ba and Se and the redox-active transition metals (Cd, Cr, Cu, Ni and V) are major constituents of firecrackers (Shirmohammadi et al. 2015; Puthussery et al. 2018; Rezaei et al. 2018; Cigánková et al. 2021; Malakootian et al. 2021). Mn and Zn also showed a moderate correlation with OPv ($r < 0.70$, $p < 0.01$), which has previously been found in several studies (Verma et al. 2010; Charrier and Anastasio 2012; Saffari et al. 2014). On the other hand, weak associations ($r < 0.50$) of OP were found with the crustal metals, like Ca, Al, K and Mg, most likely due to these metals' low water solubility (Cheung et al. 2012).

The OP of ambient particulate matter has also been employed as an indicator of aerosol toxicity in several recent investigations (Borlaza et al. 2018; Bates et al. 2019). The daily hazard index (HI) was calculated as the ratio of the experiment value of PM_{2.5} mass concentration to the reference PM_{2.5} mass concentration ($15 \mu\text{g m}^{-3}$ (WHO 2021) and $60 \mu\text{g m}^{-3}$ (NAAQS 2009)) to assess the toxicity of PM_{2.5}. Figure 7 depicts the daily variance of the measured OPv and OPm, as well as the daily Hazard Index. Figure 7 shows that the computed HI is generally greater than 1, suggesting the occurrence of adverse health effects. This comparison also demonstrates that OP of PM_{2.5} can differ from HI values. Although, the HI is usually more than 1, there was a significant fluctuation in DTT activity. During the study time, it was also observed that when the HI was high, the OP activity did not considerably rise, and vice versa. During the pre-Diwali and Diwali periods, HI levels were significantly associated with OPv values. Surprisingly, post-Diwali HI levels rise in tandem with lower observed OPv values, implying that the PM_{2.5} mass concentration alone may not be sufficient to explain PM toxicity. To better understand the health impacts of PM_{2.5}, it is necessary to assess its toxicity by identifying certain PM features (additional metric, OP) in comparison to the exposure level (mass concentration) that contributes the most to its redox activity (Borlaza et al. 2018).

3.4. SEM-EDS Analysis

The 24-h samples collected during the Diwali period, along with a blank filter, have been analysed using SEM-EDS and are shown in Fig. 8. The approach is to analyse the particles in the atmosphere during the Diwali period and identify variations in their elemental composition.

SEM analysis revealed spherical, oval, elongated, irregular, porous, agglomerates and floccules in PM_{2.5}. The difference in the shapes and sizes of the particles is mainly due to different sources of particles (Reid et al. 2005;

Wang et al. 2010; Lu et al. 2011). Figure 8 depicts the chemical composition of some of the particles based on the spot EDS spectrum for some of the particles during the Diwali period. Figure 8 (A) depicts a blank filter paper containing only C, Si, and O. Figure 8 (B) shows the SEM image along with the elemental constituents of the sample collected on the day before Diwali day (26/10/2019). It shows flaky particles rich in C, O, Na, Mg, Al, Si, S, Cl, K, Fe, Cu and Zn that may be contributed by anthropogenic sources, viz. vehicular emissions, industries, biomass combustion, road dust and soil. Figure 8 (C) shows regular crystal-like and smooth particles. The elemental analysis shows the presence of C, O, Na, Mg, Al, Si, S, Cl, K, Cu, Zn and Ba in both the particles. These particles may be contributed by industrial emission and fireworks on Diwali day (27/10/2019). In samples collected post-Diwali, irregularly shaped as well as spherical, smooth particles rich in potassium are found (Fig. 8 (D)). These particles could be tar balls that are often observed in biomass burning plumes and possess strong light absorbing properties.

The particles had a wide range of shape; many were spherical or spheroidal, while others had cleavage and crystalline characteristics. Figure 9a also shows conglomerates of tiny particles with uneven morphologies. Anthropogenic particles are formed by vapour condensation, they usually have spherical or spheroidal morphologies, while those that arise from natural sources exhibit regular morphologies with angles, cleavages, and facets [47]. Based on the intensity of the colour corresponding to each element and its relative abundance, the chemical mappings revealed that elements C, O, Al, Si, S, K and Ba were the most abundant while Na, Mg, Cl, Cu and Zn were found in comparatively lower proportions. The presence of C and K may be associated with biomass combustion derived particles, while Al and Si may be associated with alumino-silicates. Ca, Mg, and O might be linked to the presence of crystalline CaCO_3 and MgCO_3 , which could be produced from road dust and soil re-suspension. Sulphur abundance shows the presence of sulphur-bearing particles such as gypsum, as well as flaky particles containing Ca, K and Mg that could be natural or anthropogenic in origin. The presence and abundance of K, Mg, Ba, Cu and Cl suggests the contributions of fireworks. Particles emitted from fireworks are composed of these metals along with elemental carbon and secondary compounds like nitrates and organic substances. Sr, Ba and Cu compounds are used to give red, green and blue fireworks, respectively (Kulshrestha et al., 2004; Wang et al., 2007; Moreno et al., 2007). $\text{Ba}(\text{ClO}_3)_2$ is often used to impart the green colour, KNO_3 and potassium perchlorate are added to serve as an oxidizer while Mg is used as a metallic fuel and gives rise to bright electric white fireworks (Moreno et al., 2007; Wang et al., 2007) (Fig. 9B).

Conclusion

The 12-h mean $\text{PM}_{2.5}$ was found to be $263 \pm 151 \mu\text{g m}^{-3}$ during the study period which exceeded by the NAAQS limits by 4.4 times. Firework activity was also characterized by increased levels of gaseous pollutants (NO_x , SO_2 and O_3), and trace metal concentrations (Ba, Pb, Cu, Fe, Mg, K, Al and Mn). Elevated $\text{PM}_{2.5}$ - NO_x slope for fireworks including traffic emissions suggested significant contribution of fireworks. The highest levels of OP_v ($1.37 \text{ nmol min}^{-1} \text{ m}^{-3}$) and OP_m ($11.77 \text{ pmol min}^{-1} \mu\text{g}^{-1}$) were observed the morning after Diwali, exhibiting a considerable increase during the day as opposed to the night-time, indicating higher ROS activity linked with PM constituents due to fireworks that are retained in the atmosphere. Good correlations ($r > 0.700$, $p < 0.01$) were found between redox-active metals like Cr, Cu, Ni, and V and DTT activity that could be due to the ability of these metals to catalyze ROS generation in ambient air, while Ba, Be and Se in the atmosphere as major constituents of firecrackers were also strongly associated with DTT activity. A significant correlation between O_3 and PM-ROS activity was noticed which implied that the intensity of photochemical reaction including chemical species associated with fireworks is responsible for the formation of tropospheric O_3 . This was also a prominent factor influencing the formation of particulate ROS in the daytime atmosphere. Elemental mappings also revealed the abundance of C, O, Al, Si, S, K, and Ba supporting the role of fireworks. The daily OP and hazard index (HI) was compared to measure the health

effects of fine particles which suggested that measurement of PM-associated OP is needed along with mass concentration to estimate the toxicity. From a health perspective, it is required to restrict or control the fireworks activity, to mitigate human exposure that could be possible only through public awareness and emphasis on the manifestation of some effective control measures during celebrations like the Diwali festival.

Declarations

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Competing interest

The authors declare that they have no competing interests.

Author's contribution

IG has made substantial contribution to analysis and interpretation of data and writing the original draft of manuscript. PKV is involved in checking all the mathematical calculations reported in the manuscript. KMK is involved in reviewing and editing the manuscript. AL has been involved in conceptualization, data curation, funding acquisition, reviewing and editing the manuscript. All authors have read and approved the final manuscript.

Ethics approval and consent to participate

The manuscript is submitted to Springer for consideration. The submitted work is unique and has never been published elsewhere, and it has been submitted solely to the journal. The data for this manuscript is taken from other sources that were clearly cited and used with permission. The data is true and not manipulated.

Consent for publication

Authors seek their consent to Springer for publication their data prior to submitting their paper to a journal.

Availability of data and materials

The authors declare that all data supporting the findings of this study are available within the article and its supplementary information files.

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Figures

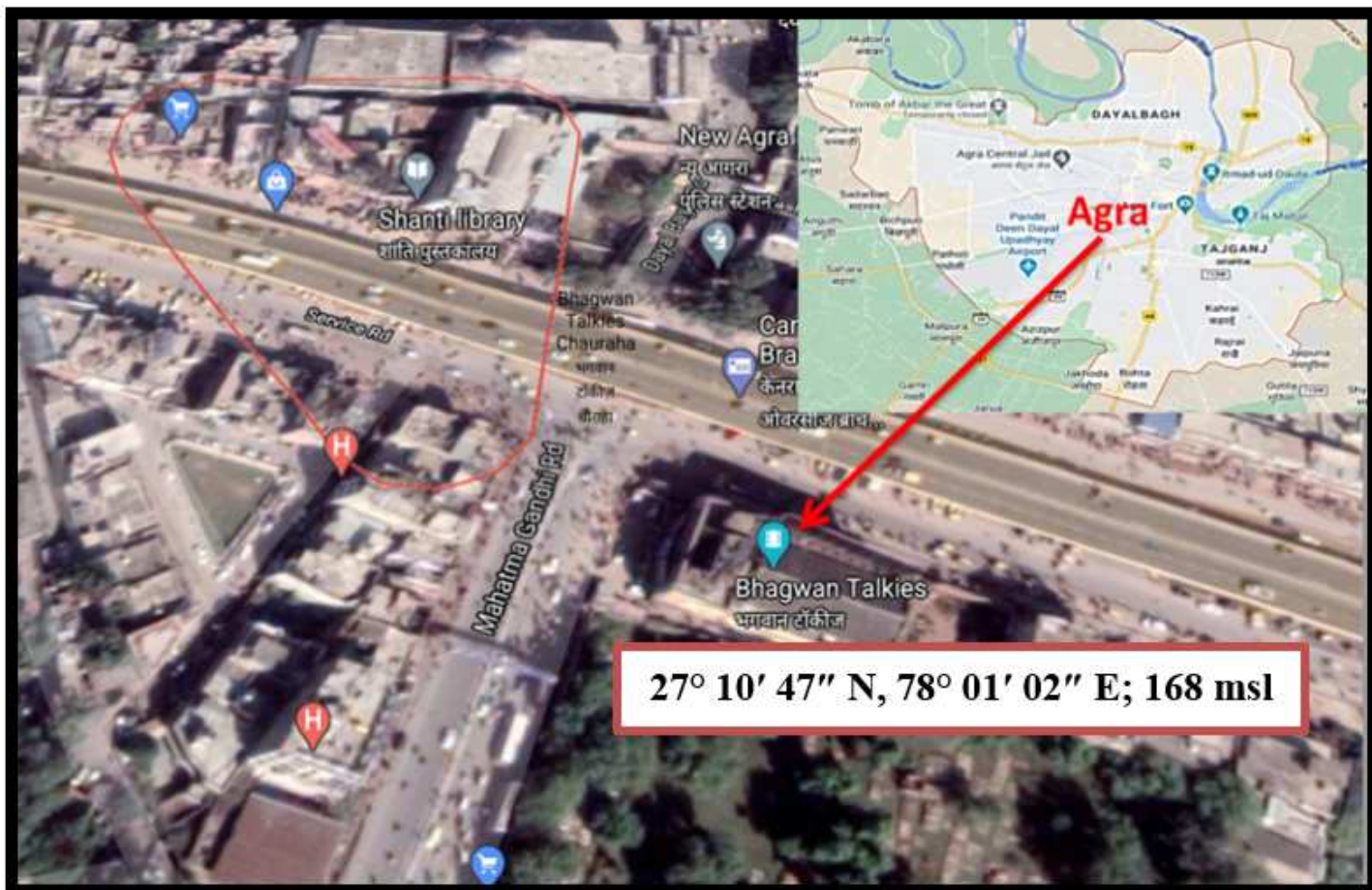


Figure 1

Location of PM_{2.5} sampling site at Agra

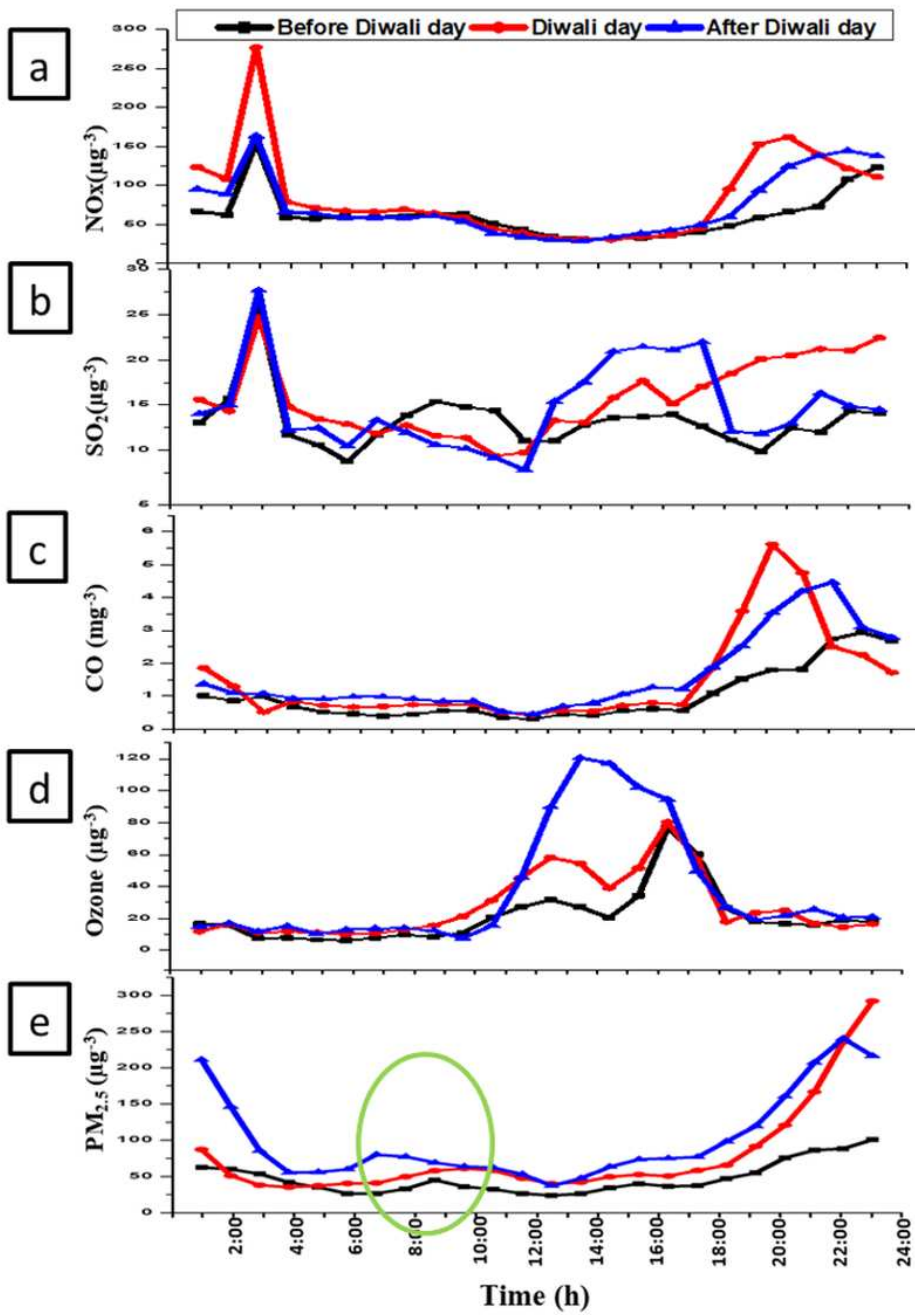


Figure 2

Diurnal profile of (a) NO_x , (b) SO_2 , (c) CO, (d) O_3 and (e) $\text{PM}_{2.5}$ mass concentration

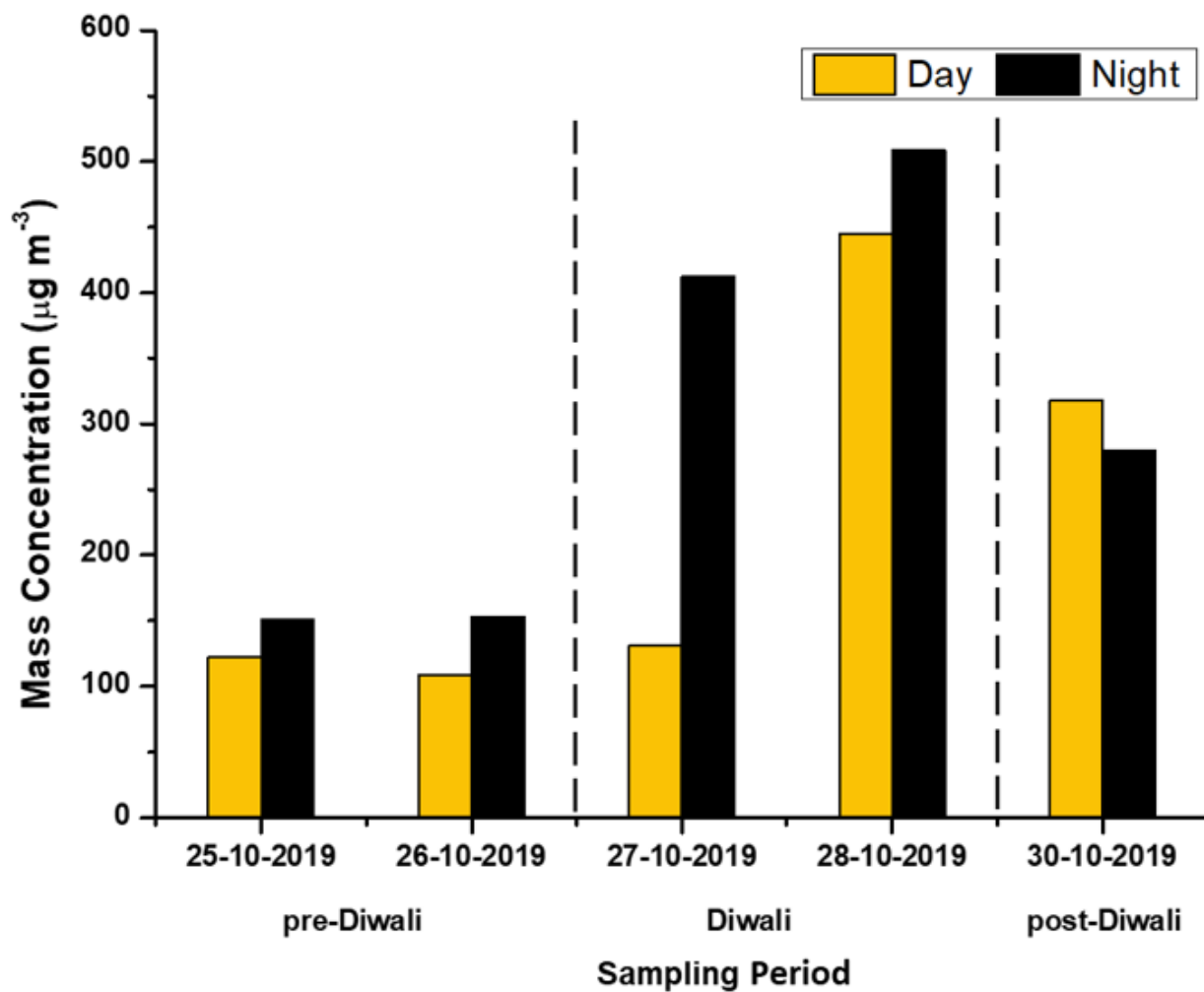


Figure 3

Day night variation of $\text{PM}_{2.5}$ mass concentrations during Diwali 2019

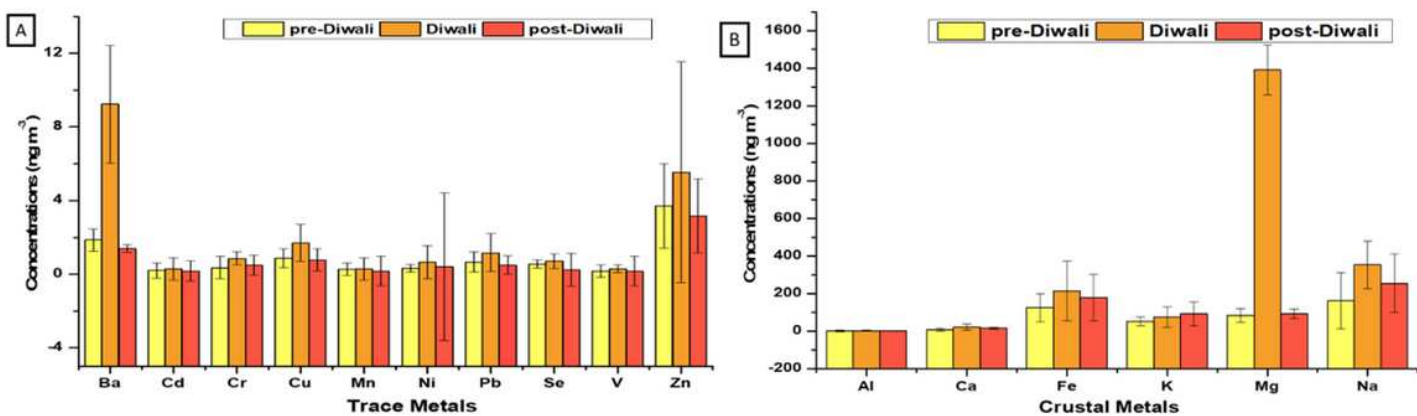


Figure 4

Variation of (a) trace and (b) crustal metal during pre-Diwali, Diwali and post-Diwali period

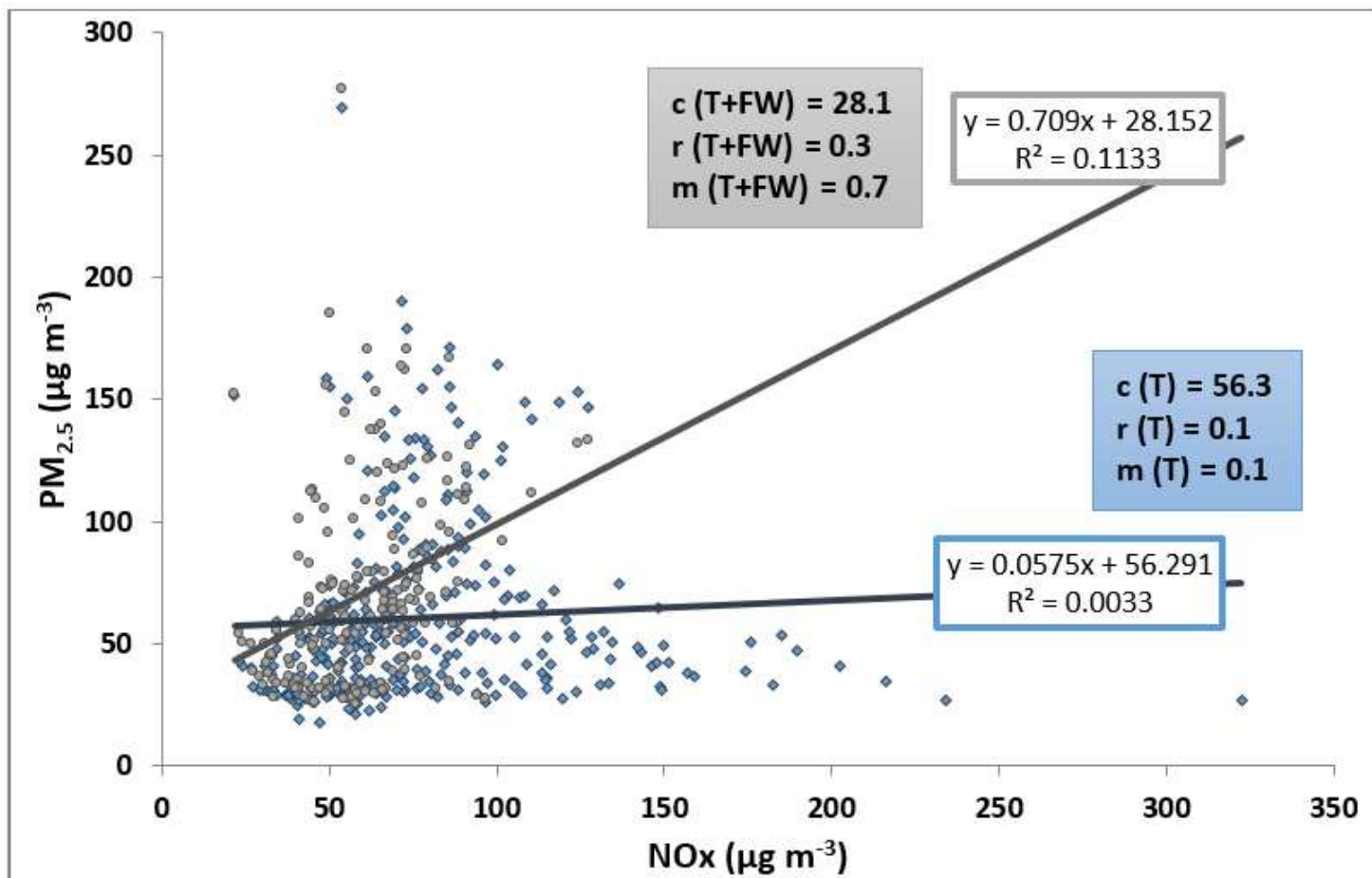


Figure 5

Regression curve of $\text{NOx-PM}_{2.5}$

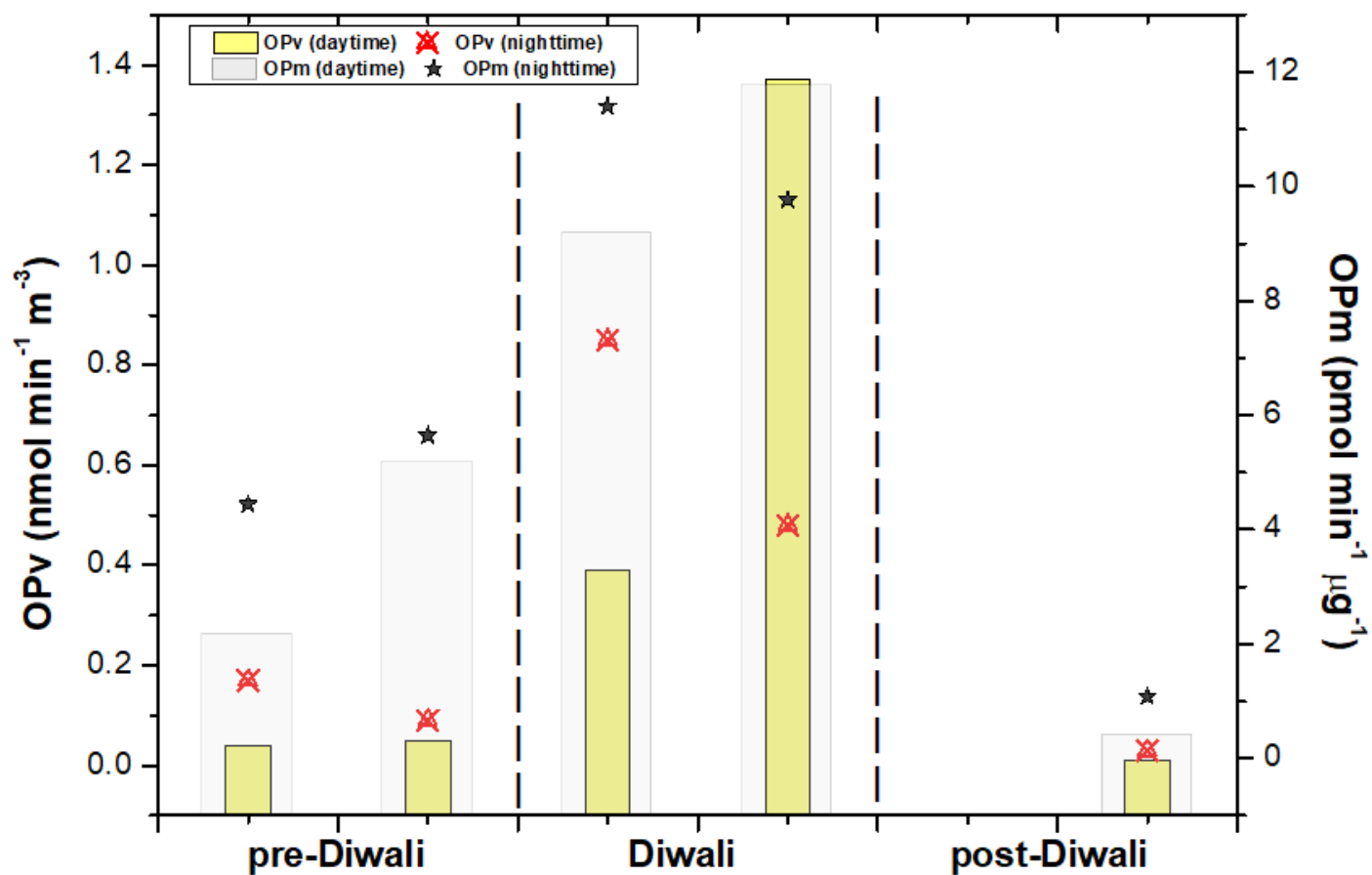


Figure 6

Diurnal Variation in OP values during pre-Diwali, Diwali and post-Diwali.

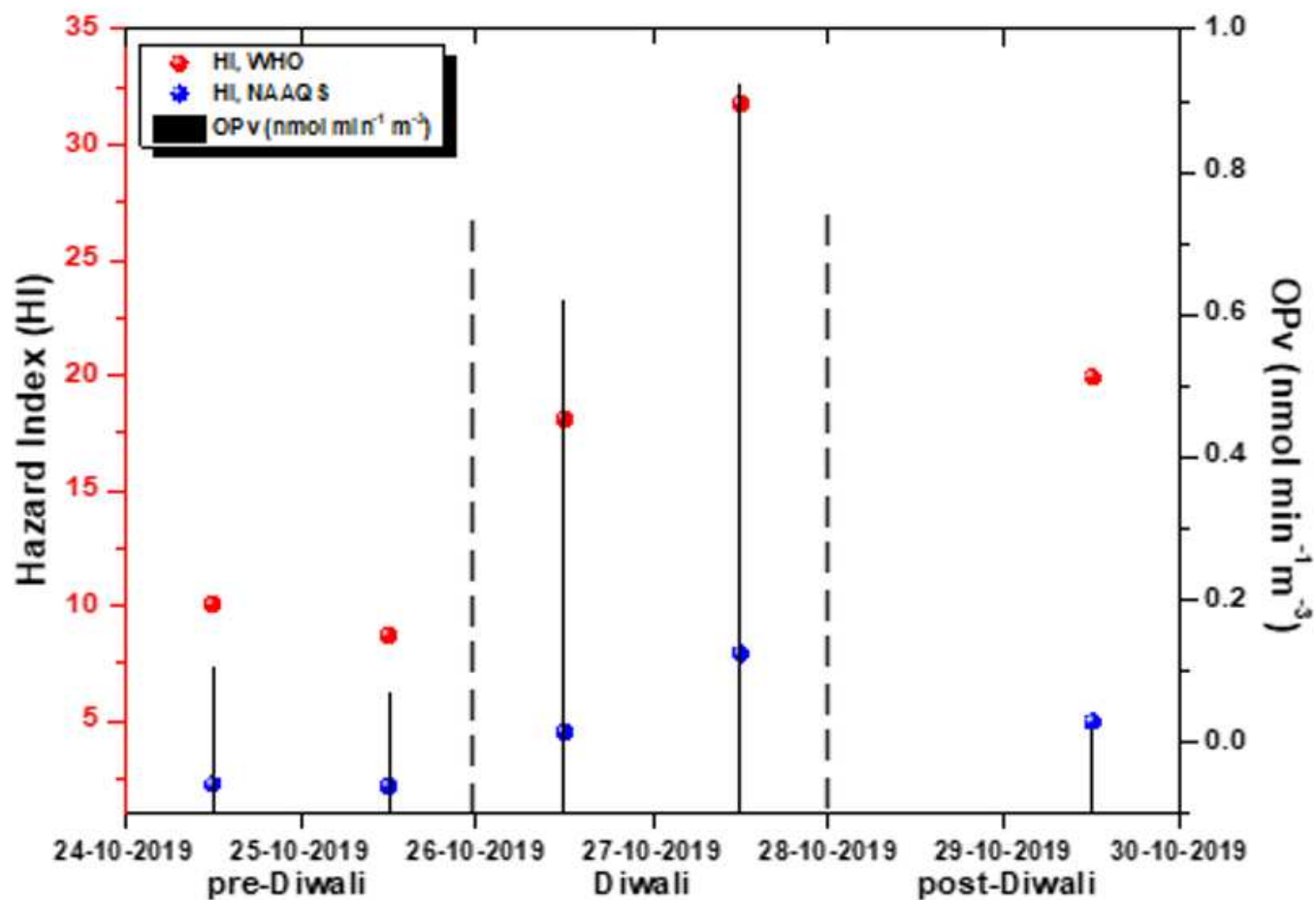


Figure 7

Daily variations in the measured DTTv and HI during Diwali period

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